Diffusion Model

Sana Alamgeer

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A Cross-Modal learning of Gaits and Balances with Joint-Aware Latent Diffusion Framework

As described in the previous section, accurate real-time assessment of human motion requires capturing the underlying CoM:BoS dynamics as they evolve during movement. However, such skeletal information cannot be directly obtained from sparse wearable sensors. This gap motivates the use of cross-modal generative learning, where the input modality consists of IMU (Inertial Measurement Unit) signals and the output is a reconstructed skeletal representation of the sensed motion. The regenerated skeletons provide explainability and richer context by making it possible to interpret postures, visualize CoM:BoS relationships, and extract biomechanical metrics during daily activities.

The proposed architecture, shown in Figure 1, adopts a multi-staged training strategy to ensure stable convergence and avoid "modal collapse". Specifically, a skeleton graph encoder is first pre-trained to establish a structured latent space, Temporal Graph Neural Network (T-GNN) encoders then learn to map wearable IMU streams into this space, and finally, a conditional diffusion model generates skeleton motion. During generation, Graph Attention Networks (GAT) preserve anatomical consistency, while cross-attention injects sensor dynamics, producing sequences that are both structurally coherent and aligned with inertial signals.

Stage 1: Skeleton Graph Encoder Pre-training As illustrated in Figure 1(a), the skeleton graph is trained first to establish a stable latent space for skeleton motion. The input skeleton sequence $\mathbf{X} \in \mathbb{R}^{T_f \times J \times d}$ is mapped into a latent $\mathbf{z}0 = \mathcal{E}\theta(\mathbf{X})$, where joints are defined as nodes and bones as edges. The latent $\mathbf{z}0$ is then perturbed into noise through the denoising diffusion probabilistic model (DDPM) [2] forward diffusion process, producing $\mathbf{z}t$. A denoiser $\varepsilon\phi(\mathbf{z}t, t)$ is trained to predict the injected noise $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, optimized using the mean squared error loss $\mathcal{L}\text{diff} = \mathbb{E}t, \mathbf{z}0, \varepsilon\big[|\varepsilon - \varepsilon\phi(\mathbf{z}_t, t)|_2^2\big]$. Through this stage, the model learns to recover clean skeleton latents from noisy ones, forming the foundation for sensor-conditioned generation.

Stage 2: Temporal GNN Encoder Pre-training: In the second stage, synchronized IMU streams $\mathbf{S} = (\mathbf{A}, \Omega)$ are used to train temporal graph neural network (T-GNN) encoders, where $\mathbf{A}, \Omega \in \mathbb{R}^{T_f \times 3}$ represent tri-axial acceleration data from the right hip and left wrist, respectively. Each T-GNN branch processes an individual sensor stream, capturing local temporal dependencies and projecting them into compact representations. The outputs of these branches are concatenated to

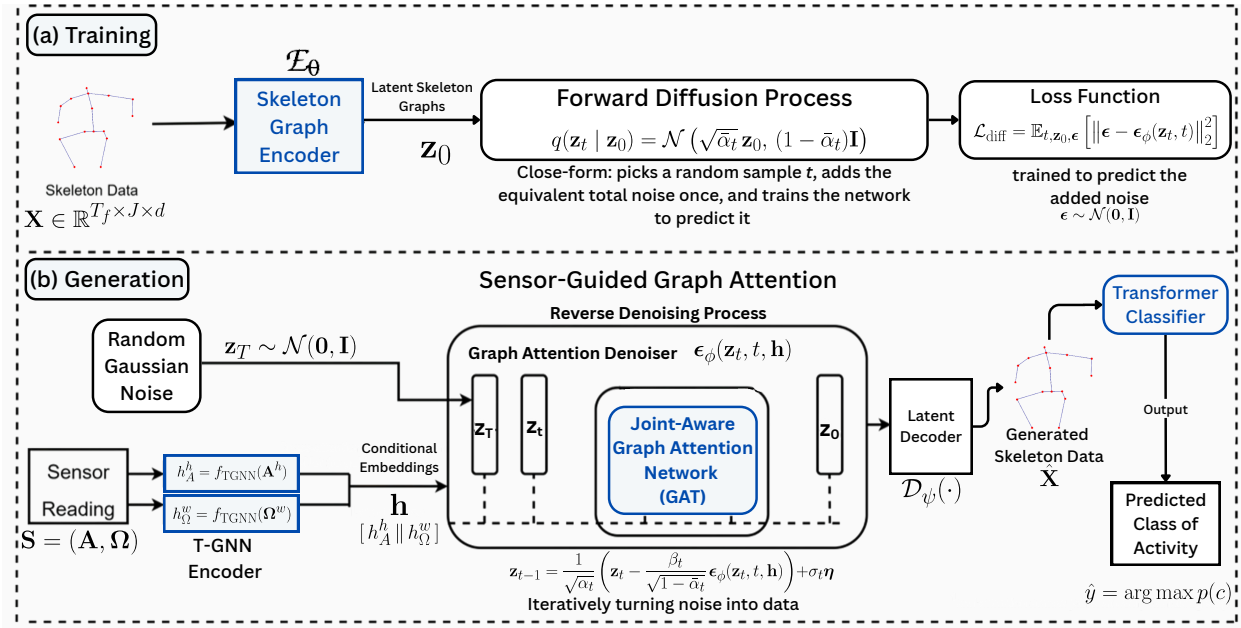

Figure 1: Illustration of the joint-aware latent diffusion framework with T-GNN encoder and GAT network.

form the sensor embedding $\mathbf{h} = f_\xi(\mathbf{S}) \in \mathbb{R}^{d_h}$, which is designed to align with the skeleton latent space established in Stage 1. Importantly, the skeleton encoder $\mathcal{E}_\theta$ trained in Stage 1 is frozen, and its latent output $\mathbf{z}0 = \mathcal{E}\theta(\mathbf{X})$ serves as the supervision target, with the objective of regressing $\mathbf{h}$ toward $\mathbf{z}0$. The training loss is defined as $\mathcal{L}_{\text{enc}} = \|\mathbf{h} - \mathbf{z}_0\|_2^2$, which enforces that sensor-derived features faithfully capture the dynamics of the skeleton space. Once trained, the T-GNN encoders provide stable conditioning embeddings that can be used to guide the diffusion model in Stage 3.

Stage 3: Diffusion Model Training In the final stage, the pre-trained skeleton encoder and T-GNN encoders are used to condition the diffusion model for skeleton generation. Starting from Gaussian noise $\mathbf{z}T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the model performs iterative reverse denoising steps using a Graph Attention Network (GAT) denoiser $\varepsilon\phi(\mathbf{z}_t, t, \mathbf{h})$, where joint features $\mathbf{z}_t$ are updated by attending over anatomically connected neighbors while being modulated by the sensor embedding $\mathbf{h}$. At each layer, the network adaptively decides how much attention each joint should pay to its connected neighbors when updating its state, thereby preserving anatomical dependencies during denoising. The reverse update is then computed as $\mathbf{z}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\phi(\mathbf{z}_t, t, \mathbf{h})\right) + \sigma_t \eta$, where $\sigma_t = \sqrt{\beta_t}$ is the noise standard deviation and $\eta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is sampled Gaussian noise. After T steps, the denoised latent $\mathbf{z}0$ is decoded into a skeleton sequence $\hat{\mathbf{X}} = \mathcal{D}\psi(\mathbf{z}0)$. The reconstructed sequence is then passed to a Transformer-based classifier [5], which outputs a probability distribution over activity classes $\mathbf{p} = [p(c_1), \ldots, p(c_K)] \in [0,1]^K$ with $\sum i = 1^K p(c_i) = 1$.

Training Datasets: We will start with SmartFallMM [4], a multimodal dataset collected by our group. The SmartFallMM dataset contains 14 action categories collected from 50 participants. It includes synchronized IMU signals from the left/right hips and wrists, along with full-body skeleton trajectories sampled at 30 FPS, providing 3D motion of 32 joints aligned with IMU data. If more data is needed,

we will source other public datasets such as the Berkeley MHAD dataset [3], which has 11 actions performed by 12 participants. The UTD-MHAD dataset [1] is another open source that can be utilized.

Potential Risks: The proposed diffusion framework offers a powerful solution for reconstructing full-body motion from minimal IMU input, but its real-world deployment could be challenging due to the sensitivity to sensor noise, which can produce biomechanically implausible joint trajectories and result in low fidelity. If this is the case, we will experiment with sensor encoders that integrate both spatial and temporal aspects of the sensor data and expand our training skeletal dataset using publicly available walking videos of older adults and extract the corresponding skeletal structures.

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